

UNIVERSITY OF TORONTO STA304H5F: Surveys, Sampling and Observational Data
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Data Collection Instructions

Part 1: Filling out other surveys.

While approving your topic, I also sent the representative (the one who submitted the form via email) the list of the course unique IDs for filling out surveys. If you did not receive the ID, then email me and I will provide you with the information.

You will receive 1% if your name is recorded for 10 or more surveys. There will be no part marks awarded for completing less than 10.

I recommend filling out more than 10 in case a group fails to submit their dataset. The code I will use to check the unique identifiers will be attached on the course webpage.

You are allowed to fill out your own survey and you can only complete any survey once. It is your responsibility to recall which surveys you have completed.

We also recommend that you write your unique identifier down on whatever tools you bring to class if you are approached to fill out a questionnaire in person.

Part 2: Dataset & Codebook

Submit your collected data set in either **.xlsx** or **.csv**. Be sure all data columns are clearly labelled with headings, and that the **data is clean and ready for analysis**. The purpose of this checkpoint is for you to be able for you to load this file in R and to be able to use it immediately instead of focusing on data cleaning. You are supposed to use the same file for your technical report.

Please ensure the following before you submit your dataset:

- Name the column including student's unique identifiers EXACTLY as "studentID". You do not need to modify the "studentID" data in case people have input weird entries; my scripts will consider of this.
 - **If you fail to name this column correctly, you will be deducted 10%. We have posted an example of the script to detect the percentage of fabrication and how we will read your .csv file.**
- Likert-scale data has been converted to numeric for analysis.
- If you have open-ended responses, you must ensure there's an extra column that includes the "codes" from thematic analysis. Instructions for how to perform thematic analysis are provided at the end of this document.

You must include a **codebook** (a separate file) that includes the description of each variable and how it was measured. Instructions of what makes a good dataset and a codebook is attached to this document after the rubric.

Rubric

(Individual) Completing Other Surveys (/1)

- The student has completed 10 or more surveys, in which the other groups submitted the data collection before the deadline. (/1)
- Note: you will NOT obtain part marks for failing to complete 10. This is to ensure most groups will have a large sample size!
- This mark is not shared amongst group members.

Presentation of Data and Proper Collection (/1)

- The group has a column named “studentID”, where more than 80% of responses correspond with a legitimate course identifier. (/0.5)
- The data looks clean and organized. The data corresponding to the column names makes sense in the given context. (/0.25)
- The group successfully gathered 50 samples. (/0.25)

Codebook (/1)

- The spelling, grammar and format of the codebook are satisfactory. (/0.5)
- The codebook explicitly defines what each value represents. (/0.5)
 - For example, if they had a variable name “gender”, they would say something like: “0 = Male, 1 = Female, 2 = Other”. Please refer to the posted examples of a codebook below.

Datasets & Codebooks Guide

This worksheet functions as a guide for preparing your dataset for analysis & things to write on a codebook.

Listed below are instances of datasets. What qualities make them good? What qualities make them challenging to analyze in code?

Dataset 1:

Gender	wellbeing	videogame_time	studying_time	positive_society
M	1	50	24	Y
F	1	10	58	Y
M	5	20	13	N
M	2	11	28	Y
F	3	38	67	Y

Dataset 2:

What's your gender?	How do you feel about this statement: video games contribute to my personal wellbeing.	How many hours do you play video games each month on average?	How many hours do you study a month on average?	Do you think video games are a positive contribution to society?
Male	Strongly agree	50 hours	24 hours	yeah
F	Strongly agree	10 h	58 h	Yes
Male	Disagree	20	13	No
M	Agree	11 hour	28 hour	ye
Female	Slightly agree	38	67	yep

Dataset 3:

Gender	Video games contributing to personal well-being	Video game play time per Month	Studying time per month	Video games positive contribution
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Male	Strongly agree	50 hours	24 hours	Yes
Female	Strongly agree	10 hours	58 hours	Yes
Male	Disagree	20 hours	13 hours	No
Male	Agree	11 hours	28 hours	Yes
Female	Slightly agree	38 hours	67 hours	Yes

Dataset 1	Dataset 2	Dataset 3
Pros: • Responses are nearly organized and likely do not need further cleaning Cons: • Columns are vague; a codebook is necessary for an explanation	Pros: • The questions are explicitly given, leaving no ambiguity Cons: • Data is not cleaned properly; will be difficult to use without further cleaning • Column names hard to work with	Pros: • Column names summarizes the data appropriately Cons: • May still need further cleaning (need to remove hours, might want to turn Likert responses into a number) • Column names hard to work with

Certain selections exhibit minimal advantages of one option over the other. For instance, using "M/F" instead of "Male/Female" as data labels doesn't change much.

Nonetheless, there are situations where the meaning behind initials or other indicators might not be evident to someone reviewing the data—especially when converting Likert scale questions into numerical values. Hence, there exist codebooks!

What is a codebook?

A codebook is an organized document that provides detailed information about the variables in the dataset. It serves as a reference guide for anyone working with the dataset. The primary purpose of a codebook is to ensure clarity, consistency, and understanding of the data, particularly when dealing with complex or unfamiliar datasets.

Codebook conventions can differ based on the organization and context. For this course's requirements, it is satisfactory to assign labels to variables and provide concise descriptions that match with the variables.

Listed below are instances of codebooks. What qualities make them good? What are the potential pitfalls of using them?

Codebook 1

Gender	The participant's gender. M = Male, F = Female, O = Other.
Wellbeing	How the participant perceives that video games contribute to my personal wellbeing. 1 = strongly agree, 4 = neither agree nor disagree, 7 = strongly disagree.
videogame_time	How much time (in hours) the participant plays video games per month on average.
studying_time	How much time (in hours) the participant studies per month on average.
positive_society	Whether the participant believes video games contribute positively to society. Y = Yes, N = No, U = Unsure.

Codebook 2

Participant Gender	M: Male, F: Female, O: Other.
Personal Wellbeing Contribution from Video Games	1: Strongly Agree, 4: Neither Agree nor Disagree, 7: Strongly Disagree.
Average Monthly Video Game Playtime (in hours)	Value must be at least 0.
Average Monthly Study Time (in hours)	Value must be at least 0.
Perceived Positive Contribution of Video Games to Society	Y: Yes, N: No, U: Unsure.

Codebook 1 Pros: • Variable names are suitable for coding (R, python, etc) for consistency. Cons: • Potential information loss when using variable names due to ambiguity (positive_society could mean a variety of different things.)	Codebook 2 Pros: • Variable names are self-explanatory. Cons: • Will need to convert the variable names to something else when using a programming language (such as R or python) due to spaces & length of variable names.
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Both codebooks meet the requirements for earning full marks in this course, though the chosen style relies on personal preference. It's essential for the codebooks to correspond with your specific data set. For instance, codebook 1 is suitable for data set 1, while codebook 2 possesses dissimilar column headers in comparison to data set 2 or 3, rendering it unsuitable for those particular data sets.

Thematic Analysis

For some reason, in the statistics curriculum students are rarely taught how to deal with qualitative data. The reason for this is because unlike quantitative data, the process for observing qualitative data is extremely subjective. Nonetheless, I find it quite a valuable skill.

- There is a great tutorial here: <https://www.scribbr.com/methodology/thematic-analysis/>
- And a popular journal article here (if you like reading academic papers): <https://doi.org/10.1191/1478088706qp063oa>

I will also outline the steps which I expect students to take. My instructions will be different than the above tutorials since I have adjusted them to suit the purposes of the course project and what I expect you to fulfill.

Step 0: determine which group members will perform the thematic analysis.

Thematic analysis shouldn't be done alone. Especially since you are in a team of at least 3 members! Other students will have different interpretations and observe different patterns. I generally recommend the whole group should do it, but pick at least three members to work together on this.

You should never do thematic analysis all in one sitting. This is because opinions/perceptions change over time. These must be performed by students willing to spend time between two days (minimum.)

Step 1: (individual) looking at the responses and pay attention to patterns.

Each member should take a good look at all the feedback, noticing patterns, but not necessarily assigning any labels first.

Here is a brief example of 5 different feedback:

- (FB1) "you know, I like the concept of group projects but my main concern is that my teammates don't want to put in the work. They just want to pass the course but I want an A+."
- (FB2) "I didn't really learn how to code that much in R. I also found it unfair that we couldn't use other programming languages, like Python, which is more commonly used in industry."
- (FB3) "projects are stressful. It is unrealistic to take 5 courses, each of them having midterms, and assignments on top of them. They stress me out, especially when group members ghost last second."
- (FB4) "one person in my group was very picky. I tried to contribute but they kept on erasing my work, telling me it wasn't good enough and it was written by AI (it wasn't) and then reported me to the instructor. I wish we had an option to write the assignments ourselves."

- (FB5) “I didn’t feel like we were given adequate preparation for coding. I had a really hard time cleaning data and performing a t-test. R has lots of dumb errors.”

Step 2: (individual) generate a list of codes and themes.

After a least an hour since step 1, re-read all of the responses again. Then, explicitly try to summarize the codes in point form. For example:

- FB1: like projects, different grade standards
- FB2: lacking in R coding, prefer to use Python
- FB3: projects are stressful, underperforming group members
- FB4: different writing standards, prefer to solo projects
- FB5: lacking in R coding

I then realise I could summarize these into the following (main) themes with codes:

- Insufficient R coding ability
 - Not enough knowledge transfer
 - Prefer to code in other languages
- Conflicts with group members
 - Different grade standards
 - Teammates don’t assist
 - Prefer to work alone

Step 3: (group) discuss your codes and themes.

Once everyone is done, compare your themes and codes. Most people should have some overlap, but you’ll identify different ideas. (I bet you can come up with a different list of themes than I did above!) Then, discuss the codes and themes until you come to agreement.

Step 4: (individual) label the codes amongst the comments.

For each comment, now assign the new code by writing them in a new column. You can assign multiple codes to one comment of course!

- FB1: Different grades standards
- FB2: Not enough knowledge transfer, Prefer to code in other languages
- FB3: Teammates don’t assist
- FB4: Different grade standards, Prefer to work alone
- FB5: Not enough knowledge transfer

Step 5: (group) compare the codes amongst each other.

For each disagreement, everyone must explain their side and try to come to an agreement. If you have too many disagreements and it cannot be resolved, then go back to the drawing board (aka go back to step 2).

This step is complete once you all agree on the codes.

Now you should be able to have a bunch of codes for each comment, which allows you to perform some quantitative analysis on qualitative data! For example, we can see the percentage of students who had conflicts with other group members or who perceive that they don't know enough R code.

When writing the report, you should announce the observed themes and implications. For example, the above comments provide us with context of why students may dislike course projects but also ways that the course instructors may improve students' learning.